

Beyond the Human Eye: A Three-Layer Verification Architecture for Large-Scale Scholarly Wikidata Item Creation

Anonymous

Anonymous affiliation

Abstract. Publishing scholarly catalogue records as Wikidata and project-Wikibase items makes identity-bearing claims public at a scale that outgrows human review. We present the verification architecture of a production pipeline that converts Hebrew-manuscript MARC records into an ontological RDF graph and then into Wikibase items, and we ask which machine judge may act in place of the curator’s eye. We compare three layers on identical fixtures: a deterministic rule engine (30 checks, including fail-closed live API rules), a generative LLM judge driven by incident-evolved rubrics, and a decision model trained with Reinforcement Learning from Contrastive Distillation (RLCD) that returns calibrated, typed answers instead of generated text. We contribute (i) the three-layer architecture with responsibilities assigned by measurement; (ii) a certification methodology—gold construction, an *unsafe-approve* metric, accuracy and determinism gates, and a tuning loop that converts every misjudgement into either a deterministic check or a rubric-encoded rule; and (iii) an empirical evaluation on 1,238 gold-labeled items across six verification channels. Our results show that (a) deterministic code, not model judgment, should own mechanical artifact detection (moving four artifact classes from the model to code removed 10 of 11 unsafe approvals); (b) LLM judges are inconsistent on identical defects, while atomic decomposition of rubric axes into typed questions makes the RLCD judge uniform; (c) calibrated confidence gating converts model uncertainty into curator review rather than silent failures; and (d) an escalation policy—RLCD judge primary, LLM as exception handler—reduces LLM workload to roughly 15% of rows while wall-clock verification drops from hours to under two minutes per 500 items. Neither judge yet passes the zero-unsafe certification gate (0.65% unsafe-approve over the full gold for the RLCD judge, 2.4% on the LLM judge’s overlapping subset), so the system keeps human approval as the final authority and assigns the LLM to confirming rows the decision model is not confident about.

1 Introduction

Cultural-heritage institutions increasingly publish their catalogues as linked data. When the target is Wikidata or a project Wikibase, item creation is the risk surface: items are public, they assert identity (“this work is by this person”),

and errors are costly to detect and undo after third parties consume them. In our domain—Hebrew manuscripts catalogued in MARC, projected through a CIDOC-CRM-based ontology into a project Wikibase and onward to public Wikidata under the WikiProject Manuscripts data model—a single run of the pipeline can propose thousands of items. Reviewing each item with the human eye does not scale.

This paper reports the verification architecture that replaced per-item human review in production, and the methodology we use to decide when a machine judge may act. The architecture has three layers:

1. a **deterministic rule engine** (30 rules: SHACL wrappers, export-quality codes, MARC grounding, duplicate probes, fail-closed live API checks);
2. a **generative LLM judge** (Gemini with an agentic tool loop; Qubrid-hosted Kimi K2.5 and DeepSeek in linear mode) driven by rubrics that encode production incidents as decision procedures; and
3. a **RLCD decision model** (TypeSafe Jev) that evaluates typed questions against the evidence and returns calibrated answers with confidence instead of generated text.

The central engineering question is not which model is “best” but *which layer may make which decision, with what evidence*. Our answer is a certification methodology: a gold set with the unsafe-approve metric, accuracy and determinism gates, and a tuning loop in which every gold miss becomes either a deterministic check or a rubric-encoded question rule. We report measurements from production data (run of 68 manuscripts, 1,238 gold-labeled verification rows, full populations for the extraction and Wikidata channels) and a double-blind-comparable evaluation design.

Contributions.

- A three-layer verification architecture for large-scale item creation, with per-layer responsibility assignment justified by measurement (Sect. ??).
- A certification methodology for machine judges: gold construction, the unsafe-approve metric, acceptance gates, double-run determinism, and a miss-to-fix tuning loop (Sect. 7).
- An empirical study comparing the LLM judge and the RLCD judge on identical fixtures against a common gold set, including an ablation isolating the value of deterministic gates (Sect. 8).
- A production escalation policy: decision model primary, LLM as exception handler, no automatic approval before certification (Sect. 9).

2 Background and Related Work

Wikidata and Wikibase quality. Wikidata’s collaborative model [1] has made item quality a research topic; quality assessments range from completeness and consistency [2] to schema-based validation. Project Wikibases inherit the same

risks at creation time, before community checking exists. SHACL [3] provides closed-shape validation and gates our upload path, but many failure modes of item creation (misattribution, wrong entity links, generated-text artifacts, identity contamination) are semantic and out of SHACL’s reach.

LLM-as-judge. Using LLMs as evaluators is established [4], with known weaknesses: position bias, verbosity bias, and—most relevant here—*self-consistency*: identical inputs can receive different verdicts [5]. Our measurements confirm the consistency problem in a verification (not preference) setting: an LLM judge flagged the same description artifact on some rows and missed it on twelve identical rows. Self-consistency voting [5] is the standard mitigation; we show that *atomic decomposition* of the rubric into independent typed questions achieves uniformity at lower cost.

Calibration and abstention. Whether language models “know what they don’t know” is studied extensively [6]. Our architecture operationalises this: the RLCD judge returns calibrated probabilities and a confidence score, and the escalation policy treats low-confidence verdicts as *review*, never as approval or rejection. This turns abstention—long identified as the safe failure mode—into the system’s primary safety mechanism.

RLCD and decision models. Reinforcement Learning from Contrastive Distillation [7] trains models to prefer contrastive chosen/rejected pairs; TypeSafe’s System One models apply the paradigm to produce a non-generative judge that evaluates typed questions (**Choice**, **Score**, binary **Noul**) against a shared state and returns probabilities over the supplied options. To our knowledge this is the first reported use of such a decision model as the primary quality judge in a semantic-web publication pipeline.

Human-in-the-loop escalation. Escalation from automated judgment to human review is a staple of deployed ML systems. Our contribution is the specific policy: the cheap, fast, calibrated judge runs on every row; the expensive generative judge runs only on rows the primary judge is not highly confident about; and certification gates decide when either judge may auto-approve (today: never).

3 System Background

The pipeline has five curator-reviewed stages: (1) AI extraction of named entities and genres from MARC records; (2) authority enrichment (Mazal/NLI, VIAF, KIMA, Wikidata reconciliation, fail-closed matching); (3) RDF graph construction over a CIDOC-CRM-based Hebrew-manuscripts ontology; (4) projection to a project Wikibase (HMO Wikibase Studio); and (5) guarded writes to public Wikidata following the WikiProject Manuscripts data model. Every curator mutation is event-versioned, and all slow work runs as heartbeated background jobs with a Postgres read-model behind a Redis-first cache stack.

Item creation happens at stages 4 and 5. Each proposed item carries a compact evidence pack: a filtered MARC slice keyed by control numbers, authority packs (VIAF, Mazal), existing-Wikidata and duplicate-probe results, per-claim provenance, and validator output. Verdicts are cached by content fingerprint, so re-verification after edits is incremental.

4 Layer 1: The Deterministic Rule Engine

The rule engine runs every deterministic check the AI judges also see, without AI. Its 30 rules fall into three families:

1. **Wrapped validators:** SHACL violation/error severities (upload-gating) and one rule per export-quality issue code (13 codes, e.g. blank-node exports, malformed labels, generic fallback descriptions), so each code is individually blockable by a curator.
2. **Deterministic checks:** class/source presence, claim datatype shapes, MARC grounding (the item’s identity control number must appear in its own local identifier; descriptions must cite the item’s own manuscript), within-run duplicate labels, language hygiene.
3. **Fail-closed API rules:** live Wikibase liveness and label drift, Wikidata QID liveness, and a duplicate-probe search. Without API access they return **error**—never **pass**.

Results follow an explicit contract (**pass/fail/not_relevant/error**); rule results are advisory except SHACL, which gates upload. Crucially for this paper, the rule engine owns the *mechanical artifact* class. During certification (Sect. 8.3) we found that encoding four mechanical checks in code—per-text parenthesis balance, identity-control-number vs. description mismatch, quote-collision garbles in Hebrew abbreviations, and truncated folio ranges—removed 10 of 11 unsafe approvals that model judgment alone produced. Deterministic code is also where the LLM judge itself is weakest: the same generated-template defect was flagged on some rows and missed on twelve identical rows by the LLM, while the rule fired uniformly.

5 Layer 2: The LLM Judge

The vendored judge agent reads the evidence pack and returns a JSON verdict with three axes (**name_ok**, **type_ok**, **role_ok**) and an **overall** of **full/partial/fail/abstain**. Rubrics are incident-evolved documents: each production incident (hallucinated institutions, invented “missing units”, cross-record contamination, wrong public-QID links) closed with a rubric rule plus a regression example. The judge runs in two modes: an agentic tool loop (Gemini) that can pull full MARC records and authority packs on demand, and a linear single-shot mode for hosted models (Qubrid Kimi K2.5, DeepSeek V4).

Two structural costs dominate. First, *consistency*: the judge re-derives its own search of the evidence at call time, so identical defects receive different verdicts across rows. Second, *latency and cost*: the judge generates a full reasoning-plus-JSON response per row (9.2–22.2s per item in our measurements; 313k input tokens for 40 items, 14m 48s wall). A 500-item verification pass takes roughly three hours.

6 Layer 3: The RLCD Decision Model

The third layer reformulates judging as *evaluation against typed questions*. The judge prompt (identical to the LLM judge’s, including rubric, prediction block, MARC context, grounding signal, and skill packs) becomes the *state*; each rubric axis becomes a typed question; and the model returns calibrated answers instead of generated text:

- **Choice** per axis: `name_ok`, `type_ok`, `role_ok` each become a Choice question over {yes, partial, no} (plus `not_applicable` for structural entities), with the rubric’s per-entity-type exceptions encoded in the question criteria.
- **Binary Noul per claim**: each present statement (P31, P1476, P2093, P50, P8189, ...) is one question—“is this statement supported by at least one evidence channel?”—answered with a probability. Questions are evaluated in parallel and independently; adding checks does not lengthen the response.
- **Atomic quality questions**: label/description quality (system-label acceptance per entity type) and a generation-artifact check read “character by character.”

Three design consequences follow. First, **overall** is *never* asked of the model: it is computed in code from the check states, so the rubric’s universal table (and its fail-over-partial tiebreaker) is deterministic. Second, *confidence gating* becomes the safety mechanism: a blocking **no** below a calibrated threshold is routed to review rather than issued as a hard fail (Sect. 8). Third, every row yields a structured *check list* (rule id, state, message, evidence, blocking flag) rendered in the same visual language as the rule-engine panel, so curators see exactly which checks failed or partially failed. Deterministic code gates (validator errors, mechanical artifacts) are applied after the model answers and can never be softened by confidence.

7 Certifying the Judges

No judge is trusted by assertion. Each judge must pass a certification before it may act on a channel:

Gold set. 1,238 rows labeled against the same evidence the judges see: full populations for the four extraction evaluators (507 rows) and the Wikidata channel

(232 rows), plus a 500-row stratified sample of the 1,949-item Wikibase projection (exportable items only; non-exportable drafts excluded). Labels cover **overall** plus the three axes, carry a labeler confidence and an escalation flag; 14 rows are flagged for curator review. A 200-row stratified subset (67/66/67 full/partial/fail) is prepared for curator countersigning; the results in this paper use the pre-countersign labels and this is disclosed as a limitation.

Unsafe-approve metric. The only error direction that can ship a bad item is a **full** verdict on a gold-**partial/fail** row. We therefore report *unsafe-approve* separately from accuracy, and its gate is zero. *Safe-miss* (a **fail** verdict on a gold-**full** row) costs review effort but cannot ship an error; we report it to separate precision loss from safety loss.

Gates. Certification requires, per channel: accuracy ≥ 0.98 against gold, unsafe-approve = 0, and double-run verdict stability ≥ 0.99 . Below the gates, judges may triage but not approve.

Tuning loop. Every certification miss is converted into a permanent fix: mechanical artifacts become deterministic checks; domain nuances (dictation notes vs. copyists; father/teacher vs. owner in provenance notes; boundary-noise spans; claim references establishing provenance) become encoded question criteria; and error-severity validator findings become code-level gates that confidence can never soften. The loop is the paper’s practical recipe: *each gold miss must end in code or in a rule, never in a prompt hope.*

8 Evaluation

All experiments use production data (one pipeline run; 68 manuscripts; 1,949 projected entities; 232 proposed Wikidata items; 507 extraction predictions). Judges see byte-identical evidence. The RLCD judge processes items concurrently (~ 5 in flight); the LLM judge runs linearly at 30 rpm.

8.1 Head-to-head on identical fixtures (E1)

On 40 items per channel judged by both systems on identical fixtures, the RLCD judge averaged 0.37–0.61 s per item (a 500-item scope completes in 39–77 s wall) versus 9.2–22.2 s per item for the LLM (40 HMO items in 14 m 48 s; a 500-item scope ≈ 3 h 05 m): a **100–250 $\times$ wall-clock advantage** for the RLCD judge. Overall agreement between the judges was 0.85–0.975 depending on channel; crucially, *every* RLCD disagreement with the LLM was in the conservative direction (route to review), and the LLM missed 12 identical template defects that the RLCD judge flagged uniformly via its atomic artifact question.

Channel	<i>n</i>	Accuracy	Unsafe-approve	Safe-miss	Cost/item
Person NER	204	0.809	0	9	\$0.21/1k
Contents NER	150	0.940	1	0	\$0.24/1k
Provenance NER	60	0.817	2	2	\$0.16/1k
Genre classifier	93	0.914	2	2	\$0.15/1k
Wikibase items	500	0.856	1	4	\$0.34/1k
Wikidata items	231	0.671	2	0	\$0.63/1k
Total	1,238	0.826	8 (0.65%)	17	\$0.42 total

Table 1. RLCD judge vs. the 1,238-row gold (full populations for the extraction and Wikidata channels; stratified 500 for the Wikibase projection). Accuracy gate is 0.98; the unsafe-approve gate is 0. Costs at the published rate of \$42 per Btok (\$0.042/Mtok input, output tokens free; console-verified: 54.2M tokens billed \$2.20), reported per 1,000 items. The full 1,238-row certification cost \$0.42.

	Model-only + code gates		Δ
Wikibase unsafe-approve (/500)	11	1	-10
Wikidata fail-class recall vs. gold	34/63	63/63	+29
Person unsafe-approve	1	0	-1

Table 2. Ablation of deterministic gates (validator severity in code; artifact checks in code; rubric nuances as encoded question criteria).

8.2 Certification against gold (E2)

Table 1 reports the post-tuning certification numbers.

For reference, the LLM judge scored on the overlapping gold subset ($n=123$) reaches 0.902 accuracy with 3 unsafe approvals (2.4% on that subset); its subsets under-represent the hardest classes, so the numbers are indicative rather than directly comparable. On the wikidata identifier-less cohort—the channel’s dominant failure class—both judges fully agree with gold (63/63) once the deterministic validator gate is in place.

8.3 Ablation: model-only vs. model + code gates (E3)

Table 2 isolates the deterministic gates’ contribution on the 500-item Wikibase channel and the Wikidata channel. Moving mechanical artifacts and validator severities from model judgment to code removed 10 of 11 unsafe approvals and made the dominant failure class (NO_IDENTIFIER, error severity) exactly reproducible (63/63).

8.4 Escalation policy (E4)

The production policy is: RLCD judge primary; LLM confirms any row that is not a high-confidence full (measured 15–25% of rows); neither judge auto-approves before certification passes. Under this policy the LLM workload drops

to roughly 15% of today’s, the LLM’s unsafe approvals are confined to the subset it confirms, and a 500-item verification completes in under two minutes instead of three hours. Because verdict caching is content-addressed, the escalation path reuses the same session, cache, and persistence machinery as the existing single-judge flow.

8.5 Determinism

A double-run of the 500-item Wikibase channel measured 0.966 overall and 0.99 per-axis verdict stability for the RLCD judge; all unstable verdicts flipped between **full** and **partial**—the review direction. This is below our 0.99 gate at the overall level and motivates majority-voting borderline axes in production (future work).

8.6 Cost

The RLCD judge costs \$0.15–0.63 per 1,000 items at the published rate of \$42 per Btok (\$0.042/Mtok input; output tokens free; console-verified: 54.2M tokens billed \$2.20). The complete 1,238-row certification cost \$0.42. The LLM judge consumed 7.8k–10.7k input and 238–422 output tokens per item; its cost is $c_{\text{item}} = t_{\text{in}} \cdot r_{\text{in}} + t_{\text{out}} \cdot r_{\text{out}}$ with the provider rate as parameter. At any plausible hosted-LLM rate the generative judge costs more per item than the RLCD judge—and is 50–250× slower. With the corrected price, the RLCD judge dominates on speed, cost, and structure simultaneously; the generative judge’s remaining advantages are its free-text audit trail and its (subset-measured) accuracy edge.

9 Production Policy

The deployed configuration follows from the measurements:

1. **RLCD judge primary** on every verification channel, emitting the structured check list per item;
2. **LLM as exception handler** for rows that are not high-confidence passes;
3. **no automatic approval** by any judge: the certification gate (zero unsafe-approve, accuracy ≥ 0.98 , determinism ≥ 0.99) must pass on the full gold first, and SHACL remains the only upload gate;
4. curators act on the union of rule-engine failures and AI check lists, with identical UI semantics across layers.

10 Limitations

The gold set was labeled by a strong LLM against the same evidence the judges see; curator countersigning is prepared (200 stratified rows) but pending, and one artifact class showed labeler disagreement that caps measurable accuracy

(≈ 0.90 reconciled on the Wikibase channel). The domain is a single manuscript pipeline; replication on other catalogues is future work. The RLCD model is English-primary and Hebrew-heavy states are not independently certified at scale. Overall determinism (0.966) is below the 0.99 gate. Finally, the LLM judge’s full-gold certification was not run (a multi-hour job); its reported numbers are subset-based.

11 Conclusion and Future Work

Large-scale scholarly item creation for Wikidata and project Wikibases can shed the per-item human eye—but only if judge authority is earned by certification, mechanical truth is pushed into code, semantic judgment is decomposed into atomic calibrated questions, and uncertainty is routed to humans rather than resolved silently. Our measurements show the RLCD decision model is the right primary judge on speed, structure, and safety-direction, with the LLM judge as a confirming exception handler; neither judge is yet certifiable for autonomous approval, and the certification methodology itself—gold, gates, and the miss-to-fix loop—is the paper’s reusable contribution. Future work includes curator-countersigned certification, self-consistency voting for borderline axes, multi-domain replication, and learned question policies that allocate model questions only where code cannot decide.

Data availability. All measurements derive from the production runs described in Sect. 3; anonymized gold labels, per-run reports, and the harness scripts are available from the authors.

References

1. D. Vrandečić, M. Krötzsch. Wikidata: A Free Collaborative Knowledgebase. *Communications of the ACM* 57(10):78–85, 2014.
2. A. Piscopo, E. Simperl. What We Talk About When We Talk About Wikidata: Quality and Use. *Proc. WWW*, 2019.
3. H. Knublauch, D. Kontokostas. Shapes Constraint Language (SHACL). W3C Recommendation, 2017.
4. L. Zheng et al. Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. *Proc. NeurIPS Datasets and Benchmarks*, 2023.
5. X. Wang et al. Self-Consistency Improves Chain of Thought Reasoning in Language Models. arXiv:2203.07181, 2022.
6. S. Kadavath et al. Language Models (Mostly) Know What They Know. arXiv:2207.05221, 2022.
7. Y. Yang et al. RLCD: Reinforcement Learning from Contrastive Distillation for Language Model Alignment. arXiv preprint, 2023.